

50 Machine Learning Questions and Answers

1. Question: What is the difference between supervised and unsupervised learning?

- o **Supervised Learning:** Uses **labeled data** to train models for predictive tasks.
 1. **Regression:** Predicts continuous values (e.g., house price, temperature).
 2. **Classification:** Predicts categories (e.g., spam/not spam, disease diagnosis).
 3. *Supervised learning predicts outcomes.*
- o **Unsupervised Learning:** Uses **unlabeled data** to discover patterns or groupings.
 1. **Clustering:** Groups similar data (e.g., customer segmentation).
 2. **Dimensionality Reduction:** Reduces features (e.g., PCA for visualization).
 3. *Unsupervised learning uncovers hidden structures.*

2. Question: What is the difference between classification and regression?

1. **Classification:** Predicts a **discrete label or category**. Evaluated using metrics such as **accuracy, precision, recall, F1-score, and ROC-AUC**.
2. **Regression:** Predicts a **continuous value**. Aims to estimate quantities such as house prices, temperature, or sales figures. Evaluated using metrics like **mean squared error (MSE), mean absolute error (MAE), and R-squared**.

3. Question: What is the bias-variance tradeoff?

1. **Bias:** Error due to overly simplistic assumptions. High bias causes **underfitting**. A model pays little attention to the training data and oversimplifies the problem.
2. **Variance:** Error due to sensitivity to small fluctuations in the training set. High variance causes **overfitting**. A model pays too much attention to the training data and captures noise.
3. The goal is to find a balance where the total error (bias + variance) is minimized.

4. **Question: How to deal with overfitting and underfitting?**

- o **Overfitting:** Model learns noise and performs poorly on new data. Caused by excessive model complexity.
 - **Combat Overfitting:** Use **regularization, cross-validation, reduce model complexity, early stopping, and collect more data.**
- o **Underfitting:** Model is too simple and performs poorly on both training and test data. Caused by insufficient model complexity.
 - **Combat Underfitting:** **Increase model complexity, add more features, reduce regularization.**

5. **Question: What is cross-validation and why is it important?**

1. **Cross-validation:** A resampling technique to assess model performance. In Fold, the data is split into subsets for training and validation.
2. **Importance:** Provides a more reliable estimate of **model generalization**, reduces **overfitting risk**, and helps in **hyperparameter tuning**.

6. **Question: What is precision, recall, and F1-score, and what is the trade-off between them?**

1. **Precision:** Proportion of predicted positives that are actual positives ().
Recall: Proportion of actual positives that are predicted as positives ().
2. **F1-score:** The harmonic mean of precision and recall, balancing both metrics.
3. **Trade-off:** Increasing precision often **decreases recall** and *vice versa*.

7. **Question: How do you choose the right evaluation metric for your problem?**

1. **Classification:** Accuracy, Precision, Recall, F1, AUC-ROC. o **Regression:** MAE, MSE, RMSE, R². o **Data Imbalance:** Use Precision, Recall, or F1 instead of Accuracy.
2. **Business Goal:** High cost of **false negatives** use **Recall**. High cost of **false positives** use **Precision**.

8. **Question: What is the difference between accuracy, precision, and recall?**

1. **Accuracy:** Out of all predictions, how many were correct? **Formula:** . Best when classes are balanced.
2. **Precision:** Out of all predicted positives, how many were actually positive? **Formula:** . Use when **false positives are costly** (e.g., spam detection).

3. **Recall (Sensitivity):** Out of all actual positives, how many did the model correctly identify? **Formula:** $\frac{TP}{TP + FN}$. Use when **false negatives are costly** (e.g., disease detection).

9. **Question: What is a Confusion Matrix and how is it used?**

1. **What it is:** Used to evaluate the performance of a classification model by comparing predicted values with actual values.
2. **Components:** True Positive (TP), False Negative (FN), False Positive (FP), True Negative (TN).
3. **Use:** Helps calculate key metrics such as Accuracy, Precision, Recall, and F1 score.

10. **Question: How do you handle missing or corrupted data in a dataset?**

1. **Deletion:** Remove rows/columns if loss is acceptable and doesn't introduce bias.
2. **Imputation:** Replace missing values with mean, median, mode, k-NN imputation, or predictive modeling.
3. **Flagging:** Add a binary indicator to signal missingness.
4. **Advanced Methods:** Use algorithms that natively handle missing data (e.g., XGBoost, LightGBM).

11. **Question: How do you handle categorical variables in machine learning?**

1. **One-hot encoding:** Creates binary columns for each category. o **Ordinal encoding:** Assigns integers if order is meaningful.
2. **Target encoding:** Replaces categories with the mean target value for that category.

12. **Question: What is feature engineering and why is it crucial?**

1. **Feature Engineering:** The process of creating, transforming, or selecting features to improve model performance.
2. **Crucial Importance:** Well-engineered features significantly boost model **accuracy, interpretability, and robustness**. Poor features lead to poor model performance regardless of the algorithm.

13. **Question: What is the difference between parametric and non-parametric models?**

1. **Parametric Models (e.g., Linear Regression):** Assume a **fixed number of parameters**. Simple, fast, less flexible, may underfit.
2. **Non-Parametric Models (e.g., Decision Trees, k-NN):** **No fixed number of parameters**; model grows with data. More flexible, but slower, needs more data, and risks overfitting.

14. **Question: What is the curse of dimensionality and how does it affect machine learning models?**

1. **Curse of Dimensionality:** Problems that arise when data has **too many features**.
2. **Effect:** Data becomes sparse, distances lose meaning, and it increases **overfitting risk** and computation time.
3. **Solution:** Use **dimensionality reduction** (e.g., PCA) or feature selection.

15. **Question: What is the purpose of regularization in machine learning?**

1. **Purpose:** Prevents **overfitting** by adding a penalty to the loss function, discouraging overly complex models.
2. **Types:** **L1 (Lasso)** can reduce coefficients to zero (**feature selection**). **L2 (Ridge)** distributes shrinkage across coefficients.

16. **Question: What are the core assumptions of linear regression?**

1. **Linearity:** Relationship is linear. o **Independence:** Observations are independent (no autocorrelation). o **Homoscedasticity:** Variance of residuals is constant.
2. **Normality of Residuals:** Residuals are normally distributed.
3. **No or Little Multicollinearity:** Independent variables are not highly correlated.

17. **Question: What is the role of activation functions in logistic regression?**

The **sigmoid activation function** is used to convert the linear output into a **probability between 0 and 1**. This introduces a **non-linear transformation** at the output layer, which is essential for handling classification tasks.

18. **Question: How do you interpret the coefficients in logistic regression?**

1. *The document provides definitions for Confusion Matrix terms instead of coefficient interpretation. The exact provided text is:*
2. **TP (True Positive):** Model correctly predicted positive class. o **FP (False Positive):** Model incorrectly predicted positive class. o **TN (True Negative):** Model correctly predicted negative class. o **FN (False Negative):** Model incorrectly predicted negative class.
3. **Use:** Helps calculate key metrics.

19. **Question: How do decision trees work in machine learning?**

1. A supervised learning model built by recursively **splitting the data** using the feature that gives the best separation (via metrics like **Gini impurity** or **information gain**).

2. **Nodes** are features/conditions, **Branches** are decision outcomes, and **Leaves** are final predictions. They can overfit, which is handled by pruning or ensemble methods.

20. Question: What is the motivation behind random forests?

1. Random forests are an **ensemble of decision trees**. The motivation is that a single decision tree is prone to **overfitting**. By building many trees on random subsets of data/features and combining their predictions, it **reduces variance**, improves generalization, and gives better performance.

21. Question: Explain the difference between bagging and boosting in ensemble methods.

1. **Bagging (e.g., Random Forest):** Models are trained **in parallel** on different subsets. It primarily reduces **variance** and is robust to overfitting.
2. **Boosting (e.g., AdaBoost, Gradient Boosting):** Models are trained **sequentially**, each correcting the errors of its predecessor. It primarily reduces **bias** and can achieve higher accuracy but is more sensitive to noisy data.

22. Question: What is the difference between hard and soft voting in ensemble methods?

1. **Hard Voting:** Each model makes a **final class prediction**, and the class with the **majority votes wins**. (No probability is used) .
2. **Soft Voting:** Each model outputs **class probabilities**. The final class is the one with the **highest average probability**. Soft voting is generally more accurate.

23. Question: What is k-nearest neighbors (k-NN) and how does it work?

1. A simple, **instance-based** algorithm. To predict a new point, it finds the **closest points** using a distance metric. For classification, it picks the majority class; for regression, it averages the neighbors' values. It requires **feature scaling** and has no training phase.

24. Question: What is k-Means and How Does It Work?

An **unsupervised clustering** algorithm that groups data into clusters based on similarity. It iteratively assigns data points to the nearest centroid and recalculates the centroids by taking the mean of all assigned points. It requires specifying beforehand.

25. Question: How do you select the best value of k in k-means clustering?

1. **Elbow Method:** Plot the sum of squared distances; the "elbow" point where the decrease sharply slows is a good value.
2. **Silhouette Score:** Measures how similar an object is to its own cluster compared to others. Higher values (up to 1) indicate better clustering.

3. **Gap Statistic:** Compares the intra-cluster variation to expected values under a null distribution.

26. **Question: What is DBSCAN clustering and how is it better than K-Means?**

1. **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** Groups data points based on **density** by connecting points with a minimum number of neighbors within a given radius (). Isolated points are labeled as **noise (outliers)**.
2. **Advantages:** It doesn't require specifying the **number of clusters** and can detect clusters of **arbitrary shapes**.

27. **Question: What is the difference between feature selection and feature extraction?**

1. **Feature Selection:** Choosing a **subset of the most relevant features** from the original dataset.
2. **Feature Extraction:** **Transforms or combines existing features** to create new ones (e.g., PCA, autoencoders).

28. **Question: How do you interpret feature importance in tree-based models?**

Typically measured by the **Reduction in Impurity** (total decrease in Gini or entropy when a feature is used for splits) and **Split Frequency**. Features with higher scores contribute more to predictions.

29. **Question: What is principal component analysis (PCA) and when should it be used?**

1. **PCA:** A **dimensionality reduction** technique that transforms features into a new set of **uncorrelated variables (principal components)**.
2. **Use:** Useful with **high-dimensional data, correlated features**, or to combat the **curse of dimensionality**. It reduces noise, redundancy, and speeds up model training.

30. **Question: What is Linear Discriminant Analysis (LDA) and when should it be used?**

1. **LDA:** A **supervised dimensionality reduction** technique primarily for **classification tasks**. It maximizes **class separability** by leveraging class information (maximizing between-class variance and minimizing within-class variance).
2. **Use:** When the goal is to improve **classification performance**.

31. **Question: How do you handle multicollinearity in regression models?**

1. **Multicollinearity:** Inflates the variance of coefficient estimates, making them unreliable.

2. **Handling: Remove correlated predictors** , apply **regularization (L1 or L2)** , or use **dimensionality reduction** like PCA.

32. **Question: How do you make your model robust to outliers?**

1. Use **Robust Algorithms** (e.g., tree-based methods). o **Outlier Removal** (e.g., IQR, Z-score). o Use **Robust Loss Functions** (e.g., Huber loss).
2. **Data Transformation** (Normalize or standardize data).

33. **Question: What is the difference between generative and discriminative models?**

1. **Generative Models (e.g., Naive Bayes, GANs):** Learn the **joint probability distribution** (). Can generate new data.
2. **Discriminative Models (e.g., Logistic Regression, SVM):** Learn the **conditional probability** (). Focus on finding boundaries between classes. Better for **classification**.

34. **Question: How do you choose which algorithm to use for a dataset?**

Selection depends on: **Problem Type, Data Size and Quality, Interpretability Needs, Computational Resources, and Domain Knowledge**. EDA and experimentation are key.

35. **Question: Explain L1 and L2 regularization and their differences.**

1. **L1 (Lasso):** Adds the **absolute value** of coefficients as a penalty. Can shrink some coefficients to **zero (feature selection)**.
2. **L2 (Ridge):** Adds the **squared value** of coefficients as a penalty. Shrinks coefficients but **does not set them to zero**.

36. **Question: What is the kernel trick in SVM and why is it useful?**

1. The kernel trick enables SVMs to handle **non-linear classification** by **implicitly mapping data to a higher-dimensional space** without explicitly calculating the transformation. It allows SVMs to find a linear decision boundary where the data becomes separable, making it effective for complex tasks.

37. **Question: What are the differences between batch, mini-batch, and stochastic gradient descent?**

1. **Batch GD:** Uses the **entire dataset** for one update. Stable but slow for large datasets.
2. **Mini-Batch GD:** Uses **small batches** (e.g., 32 or 64 samples). Balances speed and stability, and is most commonly used.

3. **Stochastic GD (SGD):** Uses **one random data point** per step. Fast, but updates are noisy and may overshoot.

38. Question: Explain how gradient descent works in machine learning

1. An optimization algorithm that minimizes a model's **loss function** by iteratively updating parameters (e.g., weights) in the direction of the **negative gradient**. The **Learning Rate** controls the step size.

39. Question: What is the Learning Rate and How Does It Affect Convergence?

1. The learning rate is a hyperparameter that controls the **size of the steps** taken during optimization. Too high can cause the model to **overshoot** the minimum; too low causes **slow convergence**.

40. Question: Explain hyperparameters and tuning methods.

1. **Hyperparameters:** Settings configured **before training** (e.g., learning rate) that control the learning process.
2. **Tuning Methods:** **Grid Search** (tests all combinations), **Random Search** (samples random combinations), and **Bayesian Optimization** (uses probabilistic models).

41. Question: How do you prevent overfitting during hyperparameter tuning?

1. Use **cross-validation** (k-fold) to reliably evaluate performance. Apply **early stopping**. Use **regularization** (L1/L2).

42. Question: When would you use grid search vs. random search?

1. **Grid Search:** Ideal for **small hyperparameter spaces** for thorough testing.
2. **Random Search:** Better for **large or high dimensional spaces**; faster and often more efficient.

43. Question: How do you evaluate a model's performance using an ROC curve?

1. The ROC curve plots the **true positive rate (recall)** against the **false positive rate**. The **AUC (Area Under Curve)** measures the model's ability to discriminate; a higher AUC indicates better performance.

44. Question: What is the silhouette score and how is it used in clustering?

1. Measures how well a data point fits within its cluster compared to other clusters. Scores range from -1 (poorly clustered) to 1 (well clustered). It is used to evaluate clustering quality and to choose the optimal number of clusters ().

45. Question: How to Select Features in High Dimensional Data?

1. **Filter methods:** Use statistical tests (e.g., correlation).

2. **Wrapper methods:** Use models to evaluate feature subsets (e.g., Recursive Feature Elimination).
3. **Embedded methods:** Feature selection happens during model training (e.g., Lasso).
4. **Dimensionality reduction:** Techniques like **PCA**.

46. **Question: What is R^2 and Adjusted R^2 in Regression, and How Do They Differ?**

1. **R^2 :** Measures the proportion of variance explained by the model.
2. **Adjusted R^2 :** A modified version that **penalizes adding irrelevant predictors**.
3. **Difference:** R^2 always increases or stays the same when features are added. Adjusted R^2 can **decrease** if added features don't improve the model.

47. **Question: What is the difference between feature selection and feature extraction?**

1. **Feature Selection:** Selects a **subset of the original features**.
2. **Feature Extraction:** Generates new features by transforming or combining existing ones (e.g., PCA).

48. **Question: What is the difference between A/B Testing and Machine Learning Model Deployment?**

1. **A/B Testing:** Comparing two versions of a feature/design to see which performs better.
2. **ML Deployment:** Putting a trained model into a production environment for automated, real-time predictions. A/B testing **validates** the model; deployment **operationalizes** it.

49. **Question: What is the Purpose of a Test Set, and How Does It Differ from a Validation Set?**

1. **Training Set:** Used to train the model.
2. **Validation Set:** Used to **tune hyperparameters** and select the best model during training.
3. **Test Set:** Used **only once** after training to evaluate the final model's performance on **unseen data**. The test set assesses **true generalization**.

50. **Question: What are the stages in a Machine Learning Project?**

1. Problem Definition.
2. Data Collection.

3. Data Cleaning & Preprocessing.
4. Exploratory Data Analysis.
5. Feature Engineering & Selection.
6. Model Training.
7. Hyperparameter Tuning & Validation.
8. Model Evaluation.
9. Deployment.
10. Monitoring & Maintenance.